

Exercise Sheet #8: AC Power and Resonances

Discussion: Dec 15, 2025

Exercise 1: Instantaneous and Average Power

If $v(t) = 160 \cos(50t)$ V and $i(t) = -33 \sin(50t - 30^\circ)$ A, calculate the instantaneous power and the average power.

Solution

First rewrite the current using

$$\sin x = \cos(x - 90^\circ) :$$

$$i(t) = -33 \cos[(50t - 30^\circ) - 90^\circ] = -33 \cos(50t - 120^\circ).$$

The instantaneous power is

$$p(t) = v(t)i(t) = 160 \cos(50t) [-33 \cos(50t - 120^\circ)] = -5280 \cos(50t) \cos(50t - 120^\circ).$$

Use the identity

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

Let $A = 50t$, $B = 50t - 120^\circ$:

$$p(t) = -5280 \cdot \frac{1}{2} [\cos(120^\circ) + \cos(100t - 120^\circ)] = -2640 [\cos 120^\circ + \cos(100t - 120^\circ)].$$

Since $\cos 120^\circ = -\frac{1}{2}$,

$$p(t) = -2640 \left[-\frac{1}{2} + \cos(100t - 120^\circ) \right] = 1320 - 2640 \cos(100t - 120^\circ).$$

Thus the **instantaneous power** is

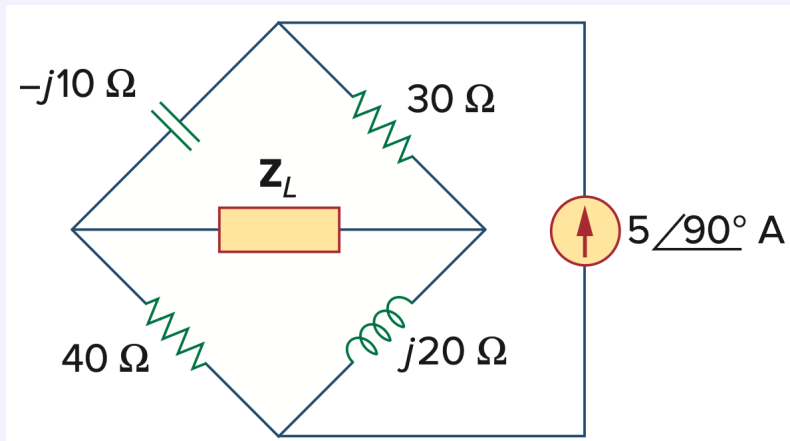
$$p(t) = 1320 + 2640 \cos(100t + 60^\circ) \text{ W}.$$

The **average power** is the DC term:

$$\bar{P} = 1320 \text{ W}.$$

Exercise 2: Maximum Average Power

Calculate the value of Z_L in the circuit below in order for Z_L to receive maximum average power. What is the maximum average power received by Z_L ?



Solution

Idea: For a network seen by a load Z_L , the maximum average power is delivered to Z_L when

$$Z_L = Z_{th}^*,$$

where Z_{th} is the Thevenin equivalent impedance seen by the load (with independent sources turned off). The maximum average power is then

$$P_{\max} = \frac{|V_{th}|^2}{8 \Re\{Z_{th}\}},$$

where V_{th} is the open-circuit (Thevenin) voltage across the load terminals.

Thevenin impedance Z_{th} : Turn off the independent current source (an ideal current source $\rightarrow$ open circuit). Then the two series branch impedances between the left and right load terminals are

$$Z_{\text{top}} = (-j10) + 30 = 30 - j10, \quad Z_{\text{bottom}} = 40 + j20.$$

These two series branches are in parallel, so

$$Z_{th} = Z_{\text{top}} \parallel Z_{\text{bottom}} = \frac{(30 - j10)(40 + j20)}{(30 - j10) + (40 + j20)}.$$

Compute numerator and denominator:

$$(30 - j10)(40 + j20) = 1400 + j200, \quad (30 - j10) + (40 + j20) = 70 + j10.$$

Thus

$$Z_{th} = \frac{1400 + j200}{70 + j10}.$$

Multiply numerator and denominator by the conjugate $70 - j10$:

$$Z_{th} = \frac{(1400 + j200)(70 - j10)}{70^2 + 10^2} = \frac{100000}{5000} = 20\ \Omega.$$

So the Thevenin impedance is purely real:

$$\boxed{Z_{th} = 20\ \Omega.}$$

Therefore the load for conjugate matching is

$$Z_L^{\text{opt}} = Z_{th}^* = 20 \, \Omega \text{ (purely real).}$$

Thevenin voltage V_{th} (open-circuit voltage across the load): With the load open, the circuit reduces to two series branches from top node T to bottom node B :

$$\text{Left branch: } Z_{L\text{total}} = (-j10) + 40 = 40 - j10, \quad \text{Right branch: } Z_{R\text{total}} = 30 + j20.$$

The independent current source is $I_s = 5\angle 90^\circ \text{ A} = j5 \text{ A}$ (upwards from bottom to top). The equivalent impedance between top and bottom is

$$Z_{\text{eq}} = (40 - j10) \parallel (30 + j20).$$

(Compute numerically)

$$Z_{\text{eq}} = 20.6 + j4.2 \, \Omega.$$

Hence the voltage between top and bottom (taking bottom as reference) is

$$V_T = I_s Z_{\text{eq}} = j5(20.6 + j4.2) = -21 + j103 \text{ V}.$$

Call the midpoint voltages (open-circuit) V_L (left mid) and V_R (right mid) measured with respect to bottom:

$$I_{\text{left}} = \frac{V_T}{40 - j10}, \quad I_{\text{right}} = \frac{V_T}{30 + j20}.$$

The mid-node voltages (voltage across the lower element in each branch) are

$$V_L = I_{\text{left}} \cdot 40, \quad V_R = I_{\text{right}} \cdot (j20).$$

Carrying out the complex arithmetic (shown below):

$$V_L = -44 + j92 \text{ V}, \quad V_R = -54 + j22 \text{ V}.$$

Thus the open-circuit (Thevenin) voltage across the load terminals (left minus right) is

$$V_{th} = V_L - V_R = (-44 + j92) - (-54 + j22) = 10 + j70 \text{ V}.$$

In polar form

$$|V_{th}| = \sqrt{10^2 + 70^2} = \sqrt{5000} = 70.7107 \text{ V}, \quad \angle V_{th} = \arctan \frac{70}{10} \approx 81.87^\circ.$$

So

$$V_{th} = 10 + j70 \text{ V} = 70.71\angle 81.87^\circ \text{ V}.$$

Maximum average power: With $Z_L = Z_{th}^* = 20 \, \Omega$ and $\Re\{Z_{th}\} = 20 \, \Omega$,

$$P_{\text{max}} = \frac{|V_{th}|^2}{8 \Re\{Z_{th}\}} = \frac{(70.71)^2}{8 \cdot 20} = \frac{5000}{160} = 31.25 \text{ W}.$$

Exercise 3: Reactive Power Compensation

An AC motor with impedance $Z_L = 4.2 + j3.6 \, \Omega$ is supplied by a 220-V, 60-Hz source.

- a) Find pf, P, and Q.

- b) Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

Solution

- a) Power factor, real power P , and reactive power Q :

The magnitude and angle of the load impedance are

$$|Z_L| = \sqrt{R^2 + X^2} = \sqrt{4.2^2 + 3.6^2} = \sqrt{30.6} = 5.531726674 \, \Omega,$$

$$\theta = \angle Z_L = \arctan \frac{X}{R} = \arctan \frac{3.6}{4.2} = 0.708626272 \, \text{rad} = 40.61^\circ.$$

The RMS current is

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{|Z_L|} = \frac{220}{5.531726674} = 39.77058393 \, \text{A}.$$

The power factor (PF) is the cosine of the impedance angle (lagging because $X > 0$):

$$\text{pf} = \cos \theta = \cos(0.708626272) = 0.7592 \quad (\text{lagging}).$$

Compute complex power using $S = V_{\text{rms}}^2 / (R - jX)^*$ equivalently

$$S = \frac{V_{\text{rms}}^2}{Z_L^*} = \frac{V_{\text{rms}}^2 (R + jX)}{R^2 + X^2}$$

so the real and reactive powers are

$$P = \frac{V_{\text{rms}}^2 R}{R^2 + X^2}, \quad Q = \frac{V_{\text{rms}}^2 X}{R^2 + X^2}.$$

Numerically,

$$R^2 + X^2 = 4.2^2 + 3.6^2 = 30.6,$$

$$P = \frac{220^2 \cdot 4.2}{30.6} \approx 6.643 \, \text{kW},$$

$$Q = \frac{220^2 \cdot 3.6}{30.6} \approx 5.695 \, \text{kVAR}.$$

- b) Capacitor required in parallel to correct PF to unity:

To correct the power factor to unity we must supply capacitive reactive power Q_C that cancels the motor's inductive reactive power Q . For a capacitor connected in parallel across the same RMS voltage V_{rms} ,

$$|Q_C| = V_{\text{rms}}^2 \omega C, \quad \omega = 2\pi f.$$

We need $|Q_C| = Q$, hence

$$C = \frac{Q}{V_{\text{rms}}^2 \omega}.$$

$$\omega = 2\pi f = 2\pi \cdot 60 = 120\pi \, \text{rad/s} \approx 376.9911184 \, \text{rad/s}$$

$$C = \frac{5694.1176471}{220^2 \cdot 376.9911184} = 312 \, \mu\text{F}$$

Exercise 4: RMS Values and Average Power

The voltage across a load and the current through it are given by $v(t) = 20 + 60 \cos(100t)$ V, $i(t) = 1 - 0.5 \sin(100t)$ A. Find:

- the RMS values of the voltage and of the current
- the average power dissipated in the load

Solution

- a) For a signal composed of a constant (DC) term plus a single sinusoid,

$$x(t) = X_{\text{DC}} + X_m \cos(\omega t + \phi),$$

the mean square (time average of $x^2(t)$) over one period is

$$\overline{x^2(t)} = X_{\text{DC}}^2 + \frac{X_m^2}{2},$$

because the cross term between DC and the sinusoid averages to zero and the sinusoid contributes $X_m^2/2$. Thus the RMS value is $X_{\text{rms}} = \sqrt{X_{\text{DC}}^2 + \frac{X_m^2}{2}}$.

Apply this to the voltage:

$$V_{\text{DC}} = 20, \quad V_m = 60$$

Hence

$$V_{\text{rms}} = \sqrt{20^2 + \frac{60^2}{2}} = \sqrt{400 + \frac{3600}{2}} = \sqrt{400 + 1800} = \sqrt{2200} = 10\sqrt{22} \text{ V}.$$

Numerically,

$$V_{\text{rms}} \approx 10\sqrt{22} \approx 46.9 \text{ V}.$$

Apply the same formula to the current. Write the current as

$$i(t) = I_{\text{DC}} + I_m \cos(100t + \phi_i).$$

Here $I_{\text{DC}} = 1$. The AC term is $-0.5 \sin(100t)$, which has amplitude $I_m = 0.5$. (The sign and phase do not affect the RMS magnitude.) Thus

$$I_{\text{rms}} = \sqrt{1^2 + \frac{(0.5)^2}{2}} = \sqrt{1 + \frac{0.25}{2}} = \sqrt{1 + 0.125} = \sqrt{1.125} = \sqrt{\frac{9}{8}} = \frac{3}{2\sqrt{2}} = \frac{3\sqrt{2}}{4} \text{ A}.$$

Numerically,

$$I_{\text{rms}} \approx \sqrt{1.125} \approx 1.061 \text{ A}.$$

- b) The instantaneous power is

$$p(t) = v(t)i(t) = (20 + 60 \cos 100t)(1 - 0.5 \sin 100t).$$

Expand:

$$p(t) = 20 \cdot 1 + 20 \cdot (-0.5 \sin 100t) + 60 \cos 100t \cdot 1 + 60 \cos 100t \cdot (-0.5 \sin 100t).$$

So

$$p(t) = 20 - 10 \sin 100t + 60 \cos 100t - 30 \cos 100t \sin 100t.$$

The average power P is the time average of $p(t)$ over one period. Every pure sinusoid (or product of sinusoids of the same frequency that reduces to a sinusoid at twice the frequency) has zero average. Thus the time average of $-10 \sin 100t$, of $60 \cos 100t$, and of $-30 \cos 100t \sin 100t$ is zero. Therefore only the DC term remains:

$$P = \overline{p(t)} = 20 \text{ W}.$$

Alternatively, using the decomposition into DC and sinusoidal parts:

$$P = V_{\text{DC}} I_{\text{DC}} + \frac{1}{2} V_m I_m \cos \varphi,$$

where φ is the phase difference between the voltage and current sinusoids. Here $V_{\text{DC}} I_{\text{DC}} = 20 \cdot 1 = 20$ and the sinusoidal part has

$$V_m = 60, \quad I_m = 0.5.$$

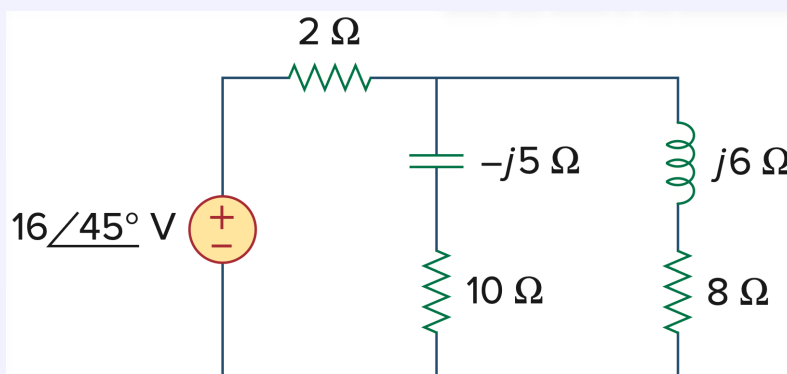
Voltage sinusoid: $60 \cos(100t)$ (phase 0). Current sinusoid: $-0.5 \sin(100t) = 0.5 \cos(100t + \frac{\pi}{2})$ (phase $+\frac{\pi}{2}$). Thus $\varphi = 0 - (+\frac{\pi}{2}) = -\frac{\pi}{2}$ and $\cos \varphi = \cos(-\frac{\pi}{2}) = 0$. Hence the sinusoidal contribution is zero and

$$P = 20 \text{ W}.$$

Exercise 5: Complex Power Calculation

For the entire circuit below, calculate:

- the complex power
- the average power delivered by the source
- the reactive power
- the apparent power
- the power factor



Solution

$$Z_1 = 10 - j5, \quad Z_2 = 8 + j6.$$

Parallel of the two branches:

$$Z_P = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} = \left(\frac{1}{10 - j5} + \frac{1}{8 + j6} \right)^{-1} = 6.1538461538 + j 0.7692307692 \, \Omega.$$

Total series impedance (including the 2Ω resistor):

$$Z_T = 2 + Z_P = 8.1538461538 + j 0.7692307692 \, \Omega.$$

Source current (phasor):

$$I_s = \frac{V_s}{Z_T} = \frac{16\angle 45^\circ}{8.1538461538 + j 0.7692307692} \approx 1.505034617 + j 1.245545890 \, \text{A}.$$

a) Complex power delivered by the source (using $S = V_s I_s^*$):

$$S_s = V_s I_s^* = 31.12 + j 2.932 \, \text{VA}.$$

b) Average (real) power delivered by source:

$$P = \Re\{S_s\} = 31.12 \, \text{W}$$

c) Reactive power:

$$Q = \Im\{S_s\} = 2.932 \, \text{VAR}$$

d) Apparent power:

$$|S_s| = \sqrt{P^2 + Q^2} = 31.26 \, \text{VA}.$$

e) Power factor (scalar) and type:

$$\text{pf} = \frac{P}{|S_s|} = \frac{31.12}{31.26} \approx 0.9956.$$

Since $Q > 0$ the net load is inductive, so the power factor is *lagging*.

Exercise 6: Series Resonance

A series RLC network has $R = 2 \, \text{k}\Omega$, $L = 40 \, \text{mH}$, and $C = 1 \, \mu\text{F}$. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

Solution

For a series RLC network with

$$R = 2 \, \text{k}\Omega, \quad L = 40 \, \text{mH}, \quad C = 1 \, \mu\text{F}$$

the series impedance is

$$\underline{Z}(j\omega) = R + j(\omega L - \frac{1}{\omega C}).$$

Resonance occurs when the net reactance is zero:

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

Substitute the values:

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}.$$

For a given angular frequency ω ,

$$X_L = \omega L, \quad X_C = \frac{1}{\omega C}, \quad X = X_L - X_C,$$

$$\underline{Z} = R + jX, \quad |\underline{Z}| = \sqrt{R^2 + X^2}, \quad \varphi = \arctan \frac{X}{R}.$$

We evaluate at $\omega = k\omega_0$ for $k = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$.

k	ω (rad/s)	X_L (Ω)	X_C (Ω)	$\underline{Z}$ (k Ω)	$ \underline{Z} $ (k Ω)	φ
0.25	1250	50	800	$2 - j0.75$	2.136	-20.6°
0.5	2500	100	400	$2 - j0.3$	2.022	-8.5°
1	5000	200	200	2	2	0°
2	10000	400	100	$2 + j0.3$	2.022	$+8.5^\circ$
4	20000	800	50	$2 + j0.75$	2.136	$+20.6^\circ$

Observations:

- At resonance ($k = 1$) the inductive and capacitive reactances cancel, leaving purely resistive impedance $Z = R = 2000 \Omega$.
- The network is capacitive ($X < 0$) for frequencies below resonance and inductive ($X > 0$) for frequencies above resonance.
- The magnitudes at pairs $k = \frac{1}{4}$ and $k = 4$ are equal (symmetry), and likewise for $k = \frac{1}{2}$ and $k = 2$.

Exercise 7: Parallel Resonance

Rework Exercise 6 if the elements are connected in parallel.

Solution

For the parallel connection the total admittance is

$$\underline{Y}(j\omega) = \underbrace{\frac{1}{R}}_G + j \left(\underbrace{\omega C}_{B_C} + \underbrace{\left(-\frac{1}{\omega L}\right)}_{B_L} \right) = G + jB(\omega).$$

The impedance is the reciprocal of the admittance:

$$\underline{Z}(\omega) = \frac{1}{\underline{Y}(\omega)} = \frac{1}{G + jB} = \frac{G - jB}{G^2 + B^2}.$$

Its magnitude and phase are

$$|\underline{Z}| = \frac{1}{\sqrt{G^2 + B^2}}, \quad \varphi = \text{atan2}(\Im\{Z\}, \Re\{Z\}) = \arctan \frac{-B}{G}.$$

Parallel resonance (net susceptance zero) occurs at the same angular frequency as series resonance:

$$B(\omega_0) = 0 \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}.$$

With the given values

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}.$$

Also

$$G = \frac{1}{R} = \frac{1}{2000} = 5 \times 10^{-4} \text{ S}.$$

At resonance $B(\omega_0) = 0$, so

$$Y(\omega_0) = G \quad \Rightarrow \quad Z(\omega_0) = \frac{1}{G} = R = 2000 \Omega.$$

We compute for each ratio k :

$$\omega = k\omega_0, \quad B = \omega C - \frac{1}{\omega L}, \quad \underline{Z} = \frac{G - jB}{G^2 + B^2}.$$

k	ω (rad/s)	B_C (m Ω)	B_L (m Ω)	$\underline{Z}$ (Ω)	$ \underline{Z} $ (Ω)	φ
0.25	1250	1.25	20	$1.42 + j 53.30$	53	$+88.5^\circ$
0.5	2500	2.5	10	$8.85 + j 132.74$	133	$+86.2^\circ$
1	5000	5	5	2000	2000	0°
2	10000	10	2.5	$8.85 - j 132.74$	133	-86.2°
4	20000	20	1.25	$1.42 - j 53.30$	53	-88.5°

Notes on the results:

- At resonance the susceptances of the inductor and capacitor cancel and the parallel branch behaves as a pure conductance G , giving $Z(\omega_0) = R$.
- For frequencies below resonance ($k < 1$) the net susceptance B is negative (inductive behavior), producing an impedance with a large positive imaginary part (inductive reactance).
- For frequencies above resonance ($k > 1$) B is positive (capacitive behavior), producing impedance with a large negative imaginary part (capacitive reactance).
- The impedances occur in conjugate pairs for reciprocal frequency ratios (e.g. $k = 0.25$ and $k = 4$; $k = 0.5$ and $k = 2$).